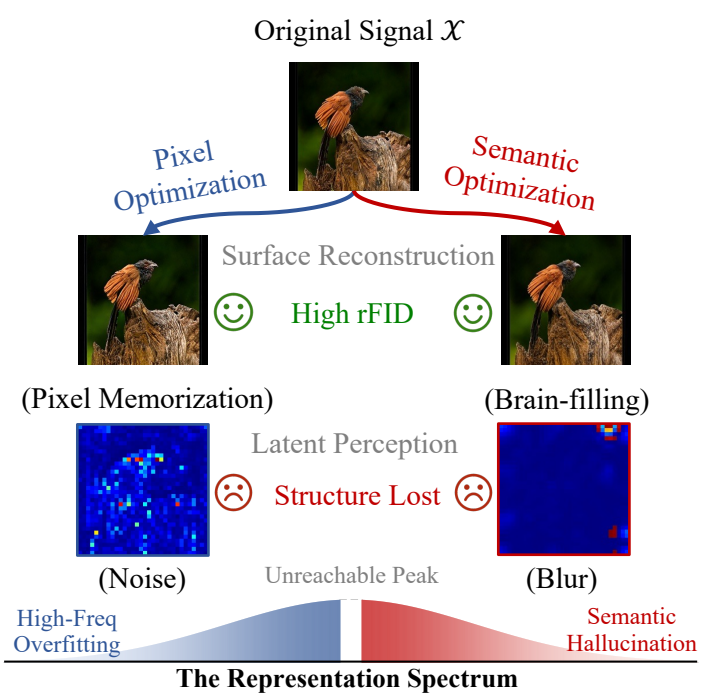
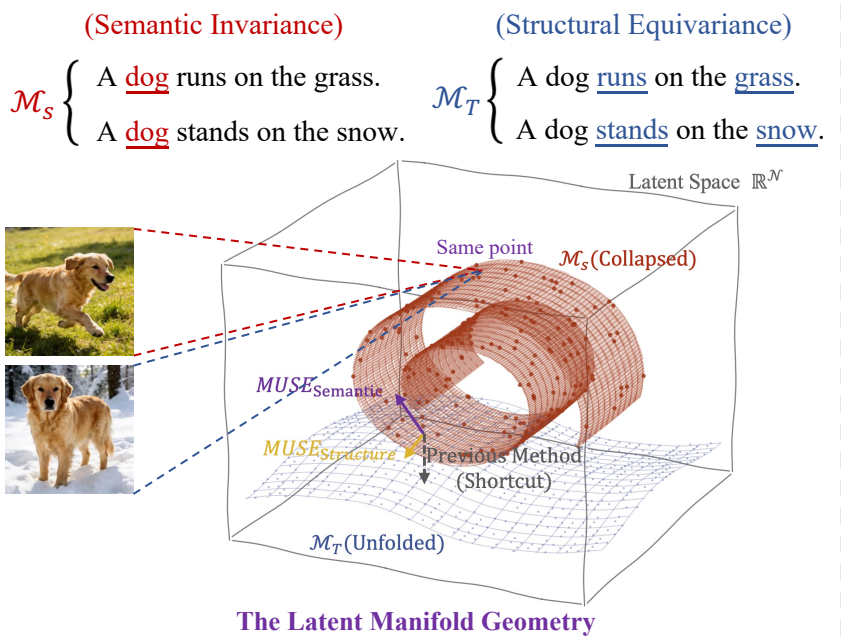
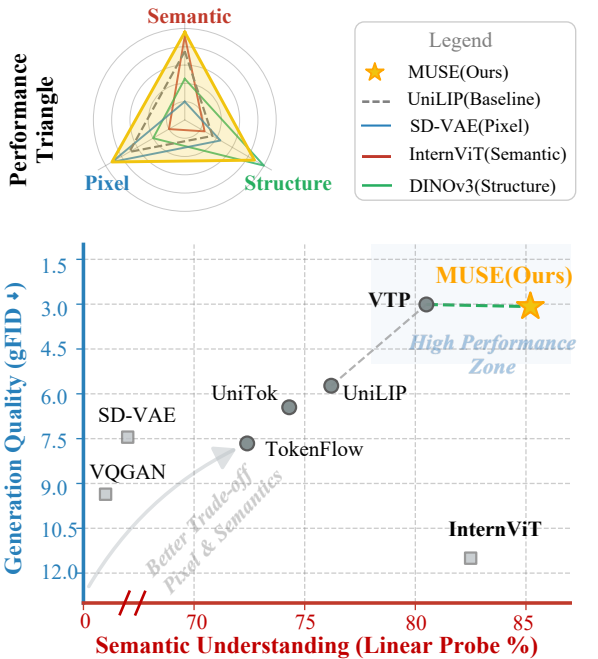

Original Signal $\mathcal{X}$
Pixel Optimization
Semantic Optimization
Surface Reconstruction
High rFID
(Pixel Memorization)
(Brain-filling)
Latent Perception
Structure Lost
(Noise)
(Blur)
Unreachable Peak
High-Freq Overfitting
Semantic Hallucination
The Representation Spectrum


(Semantic Invariance)
(Structural Equivariance)
$\mathcal{M}_s$ A dog runs on the grass. A dog stands on the snow.
$\mathcal{M}_T$ A dog runs on the grass. A dog stands on the snow.
Latent Space $\mathbb{R}^{\mathcal{N}}$
Same point
$\mathcal{M}_s$(Collapsed)
$MUSE_{Semantic}$
$MUSE_{Structure}$
Previous Method (Shortcut)
$\mathcal{M}_T$(Unfolded)
The Latent Manifold Geometry


Performance Triangle
Semantic
Pixel
Structure
Legend
MUSE(Ours)
UniLIP(Baseline)
SD-VAE(Pixel)
InternViT(Semantic)
DINOv3(Structure)
Generation Quality (gFID ↓)
1.5
3.0
4.5
6.0
7.5
9.0
10.5
12.0
MUSE(Ours)
VTP
High Performance Zone
UniTok
UniLIP
SD-VAE
TokenFlow
VQGAN
InternViT
Better Trade-off Pixel & Semantics
0
70
75
80
85
Semantic Understanding (Linear Probe %)